

Quantum 2 HW4

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Technion

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1 The Optical Theorem

1.1 Why S is unitary

In an elastic scattering the kinetic energy of the incident particle is conserved. Therefore applying the operator S to a state does not change its magnitude in the energy space, therefore it is unitary.

1.2 Show VS

We know that

$$H_0 |\psi_i\rangle = E |\psi_i\rangle \quad (1)$$

$$H |\psi_o\rangle = E |\psi_o\rangle \quad (2)$$

inserting the full form of the hamiltonian

$$(H_0 + V) |\psi_o\rangle = E |\psi_o\rangle \quad (3)$$

subtracting a zero on the right hand side

$$(H_0 + V) |\psi_o\rangle = E |\psi_o\rangle - 0 \quad (4)$$

$$(H_0 + V) |\psi_o\rangle = E |\psi_o\rangle - (E - H_0) |\psi_i\rangle \quad (5)$$

$$(H_0 + V) S |\psi_i\rangle = ES |\psi_i\rangle - (E - H_0) |\psi_i\rangle \quad (6)$$

$$VS |\psi_i\rangle = ES |\psi_i\rangle - (E - H_0) |\psi_i\rangle - H_0 S |\psi_i\rangle \quad (7)$$

$$VS = ES - E + H_0 - H_0 S \quad (8)$$

$$VS = (E - H_0) (S - \mathbb{1}) \quad (9)$$

in the limit $v \rightarrow 0$ S is the identity, since both sides of the equation must equal 0 always.

1.3 Resolvent

If we apply G_0 to both sides

$$G_0 VS = G_0 (E - H_0) (S - \mathbb{1}) \implies G_0 VS = S - \mathbb{1} \implies S = \mathbb{1} + G_0 VS \quad (10)$$

1.4 Projection

We'll apply the above equation to the state $|k\rangle$

$$S|k\rangle = \mathbb{1}|k\rangle + G_0VS|k\rangle \quad (11)$$

$$|\psi_o\rangle = |k\rangle + G_0V|\psi_o\rangle \quad (12)$$

1.5 Collision Operator

directly

$$S = \mathbb{1} + G_0VS \implies S = \mathbb{1} + G_0T \quad (13)$$

1.6 Applying to the State

applying state $|k\rangle$

$$S|k\rangle = \mathbb{1}|k\rangle + G_0T|k\rangle \quad (14)$$

$$|\psi_o\rangle = |k\rangle + G_0T|k\rangle \quad (15)$$

This is another representation of the Lippman Schwinger equation.

1.7 Projection Again

the form we found for $\psi_o(r)$ is

$$\psi_o(r) = \frac{1}{(2\pi)^{3/2}} \left(e^{i\mathbf{k}\cdot\mathbf{r}} + \frac{e^{ikr}}{r} f(k, k') \right) \quad (16)$$

where

$$f(k, k') = -\frac{1}{4\pi} \frac{2m}{\hbar^2} (2\pi)^3 \langle k'|V|\psi_o\rangle \quad (17)$$

transforming using our known operators

$$f(k, k') = -\frac{1}{4\pi} \frac{2m}{\hbar^2} (2\pi)^3 \langle k'|VS|k\rangle \quad (18)$$

$$= -\frac{1}{4\pi} \frac{2m}{\hbar^2} (2\pi)^3 \langle k'|T|k\rangle \quad (19)$$

1.8 Showing Things

directly

$$V|\psi_o\rangle = VS|k\rangle = T|k\rangle \quad (20)$$

using LS we found earlier

$$V|\psi_o\rangle = V|k\rangle + VG_0T|k\rangle \quad (21)$$

$$T|k\rangle = V|k\rangle + VG_0T|k\rangle \quad (22)$$

$$T = V + VG_0T \quad (23)$$

1.9 Showing More Things

$$\Im(\langle k|T|k\rangle) = \Im(\langle k|VS|k\rangle) \quad (24)$$

$$= \Im(\langle k|V|\psi_o\rangle) \quad (25)$$

$$= \Im(\langle\psi_o|V|\psi_o\rangle - \langle\psi_o|VG_0V|\psi_o\rangle) \quad (26)$$

$$= \Im(\langle\psi_o|V|\psi_o\rangle) - \Im(\langle\psi_o|VG_0V|\psi_o\rangle) \quad (27)$$

$$= \Im(\langle\psi_o|V|\psi_o\rangle) - \Im(\langle\psi_o|V(\Re G_0 + i\Im G_0)V|\psi_o\rangle) \quad (28)$$

$$= -\Im(\langle\psi_o|V\Re G_0V|\psi_o\rangle) + \Im(i\langle\psi_o|V\Im G_0V|\psi_o\rangle) \quad (29)$$

$$= -\Im(i\langle\psi_o|V\Im G_0V|\psi_o\rangle) \quad (30)$$

$$= -\Im\left(i\langle\psi_o|V\Im\left(\frac{1}{E-H_0+o\varepsilon}\right)V|\psi_o\rangle\right) \quad (31)$$

$$= -\pi\Im(i\langle\psi_o|V\delta(E-H_0)V|\psi_o\rangle) \quad (32)$$

$$= -\pi\langle\psi_o|V\delta(E-H_0)V|\psi_o\rangle \quad (33)$$

1.10 Conclude

$$-\pi\langle\psi_o|V\delta(E-H_0)V|\psi_o\rangle = -\pi\langle k|S^\dagger V\delta(E-H_0)VS|k\rangle \quad (34)$$

$$= -\pi\langle k|T^\dagger\delta(E-H_0)T|k\rangle \quad (35)$$

inserting an identity

$$-\pi\langle k|T^\dagger\delta(E-H_0)T|k\rangle = -\pi\int dk'\langle k|T^\dagger\delta(E-H_0)|k'\rangle\langle k'|T|k\rangle \quad (36)$$

next we notice that since the momentum is $P = \hbar k$ the hamiltonian is $H_0 = \frac{\hbar^2 k^2}{2m}$

$$-\pi\int dk'\langle k|T^\dagger\delta(E-H_0)|k'\rangle\langle k'|T|k\rangle = -\pi\int dk'\delta\left(E - \frac{\hbar^2 k'^2}{2m}\right)\langle k|T^\dagger|k'\rangle\langle k'|T|k\rangle \quad (37)$$

$$= -\pi\int dk'\delta\left(E - \frac{\hbar^2 k'^2}{2m}\right)|\langle k'|T|k\rangle|^2 \quad (38)$$

moving integration parameter to the energy we notice $\frac{dE}{dk} = \frac{\hbar^2 k}{m}$ therefore $\frac{dk}{dE} = \frac{m}{\hbar^2 k}$ and remembering we're now integrating over a sphere in energy phase space

$$-\pi\int dk'\delta\left(E - \frac{\hbar^2 k'^2}{2m}\right)|\langle k'|T|k\rangle|^2 = -\pi\int dE d\Omega k^2 \frac{m}{\hbar^2 k} \delta\left(E - \frac{\hbar^2 k'^2}{2m}\right)|\langle k'|T|k\rangle|^2 \quad (39)$$

$$= -\frac{\pi m}{\hbar^2 k} \int dE d\Omega k^2 \delta\left(E - \frac{\hbar^2 k'^2}{2m}\right)|\langle k'|T|k\rangle|^2 \quad (40)$$

$$= -\frac{\pi m k}{\hbar^2} \int d\Omega |\langle k'|T|k\rangle|^2 \quad (41)$$

1.11 Prove Optical Theorem

In subsection 7 we found

$$f(k, k') = -\frac{1}{4\pi} \frac{2m}{\hbar^2} (2\pi)^3 \langle k' | T | k \rangle \quad (42)$$

therefore

$$\Im(f(\theta = 0)) = \Im(f(k, k)) \quad (43)$$

$$= -\frac{1}{4\pi} \frac{2m}{\hbar^2} (2\pi)^3 \Im(\langle k | T | k \rangle) \quad (44)$$

$$= -\frac{1}{4\pi} \frac{2m}{\hbar^2} (2\pi)^3 \left(-\frac{\pi m k}{\hbar^2} \int d\Omega |\langle k' | T | k \rangle|^2 \right) \quad (45)$$

$$= \frac{k}{2\hbar^4} (2\pi)^3 \int d\Omega |\langle k' | T | k \rangle|^2 \quad (46)$$

$$= \frac{k}{2\hbar^4} (2\pi)^3 \left(-\frac{1}{4\pi} \frac{2m}{\hbar^2} (2\pi)^3 \right)^{-2} \int d\Omega |f(k, k)| \quad (47)$$

$$= \frac{k}{4\pi} \int d\Omega |f(k, k)| \quad (48)$$

using the definition for the differential cross section $\frac{d\sigma}{d\Omega} = |f(k, k')|^2$

$$\Im(f(\theta = 0)) = \frac{k}{4\pi} \int d\Omega \frac{d\sigma}{d\Omega} \quad (49)$$

$$= \frac{k}{4\pi} \sigma \quad (50)$$

2 Quantization of the Field

1. Express $\hat{H}_F$ in terms of $\hat{a}$ and $\hat{a}^\dagger$ and the number operator $\hat{N} = \hat{a}^\dagger \hat{a}$.
2. Calculate the expectation values $\langle \hat{E}(\mathbf{r}) \rangle_n$, $\langle \hat{B}(\mathbf{r}) \rangle_n$, $\langle \hat{H}_F \rangle$ in state $|n\rangle$.

2.1 Field Hamiltonian

From the definitions for the photon creation and annihilation operators we can find expressions for $\hat{\chi}$ and $\hat{\Pi}$ using the operators:

$$\hat{\chi} = \frac{1}{\sqrt{2}} (\hat{a} + \hat{a}^\dagger) \quad \hat{\Pi} = \frac{1}{\sqrt{2}i} (\hat{a} - \hat{a}^\dagger) \quad (51)$$

therefore

$$\hat{H}_F = \frac{\hbar\omega}{2} (\hat{\chi}^2 + \hat{\Pi}^2) \quad (52)$$

$$= \frac{\hbar\omega}{4} ((\hat{a} + \hat{a}^\dagger)^2 - (\hat{a} - \hat{a}^\dagger)^2) \quad (53)$$

$$= \frac{\hbar\omega}{4} (\hat{a}^2 + \hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a} + \hat{a}^{\dagger 2} - \hat{a}^2 + \hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a} - \hat{a}^{\dagger 2}) \quad (54)$$

$$= \frac{\hbar\omega}{2} (\hat{a}\hat{a}^\dagger + \hat{N}) \quad (55)$$

$$= \frac{\hbar\omega}{2} (\hat{a}^\dagger\hat{a} - \hat{a}^\dagger\hat{a} + \hat{a}\hat{a}^\dagger + \hat{N}) \quad (56)$$

$$= \frac{\hbar\omega}{2} (2\hat{N} + [\hat{a}, \hat{a}^\dagger]) \quad (57)$$

$$= \frac{\hbar\omega}{2} (2\hat{N} + 1) \quad (58)$$

$$= \hbar\omega \left(\hat{N} + \frac{1}{2} \right) \quad (59)$$

2.2 Expectation Values

First we notice

$$\langle \hat{E}(\mathbf{r}) \rangle_n = \langle n | \hat{E} | n \rangle \quad (60)$$

which if we look at the definition of the creation and annihilation operators which define $\hat{E}$, the result of the above equation is always zero, therefore

$$\langle \hat{E}(\mathbf{r}) \rangle_n = 0 \quad (61)$$

and the same holds for $\langle \hat{B}(\mathbf{r}) \rangle_n$. For $\hat{H}_F$:

$$\langle n | \hat{H}_F | n \rangle = \hbar\omega \langle n | \left(\hat{N} + \frac{1}{2} \right) | n \rangle = \hbar\omega \left(n + \frac{1}{2} \right) \quad (62)$$